



ARIN PUBLISHING ACADEMY

Research Writing, Publishing, and Policy Communication for African Scholars

MODULE 1: ACADEMIC WRITING MASTERY

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Trainer



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Dr. Maureen Ngesa is a behavioural systems and programme design leader with a background in clinical psychology and mental health and psychosocial research..

Her expertise in behavioural science, systems thinking, and multi-stakeholder engagement brings a rigorous, evidence-grounded lens to research communication, knowledge translation, and capacity building, skills directly applicable to supporting African scholars in producing and disseminating high-quality, policy- relevant research.

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Pre-Training Evaluation



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But First...

Meet your fellow Participants

- Name/ Country
- Field of study/profession
- My research interests
- My preferred name is....

Module 1- Goal and Outcomes

Goal: To equip participants with the skills required to develop high-quality scholarly manuscripts that meet international publishing standards.

Learning Outcomes

1. Structure a publishable academic manuscript
2. Develop clear research arguments
3. Write compelling abstracts and introductions
4. Present findings effectively
5. Select appropriate journals for publication
6. Improve manuscript quality before submission

Session 1: Foundations of Academic Writing

Topics

- Characteristics of scholarly writing
- Research-to-publication pathway
- Understanding journal article structures
- IMRAD format (Introduction, Methods, Results and Discussion)

Learning Objectives

By the end of this session, participants will be able to:

- Describe the characteristics of scholarly writing.
- Explain the research-to-publication pathway.
- Identify the major sections of a scientific journal article.
- Apply the IMRAD structure to organize a manuscript effectively.

Characteristics of scholarly writing

What is Scholarly Writing?

Scholarly writing is a formal, evidence-based style of communication used to disseminate research findings to the scientific community. It is designed to be clear, objective, accurate, and reproducible.

Unlike general writing, scholarly writing prioritizes evidence over opinion and follows internationally accepted conventions for reporting research.

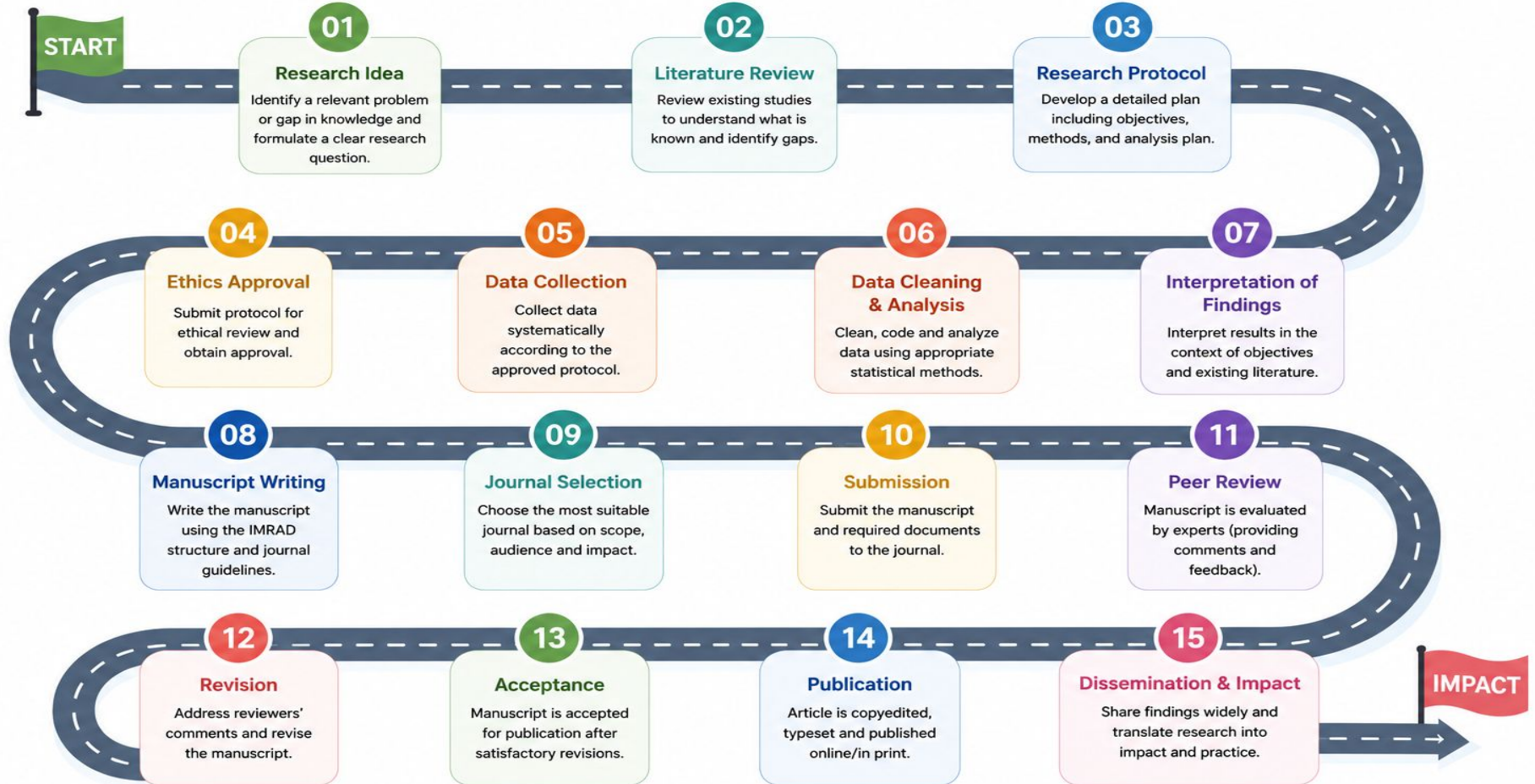
Common Mistakes in Scholarly Writing

- Writing long, complicated sentences.
- Making unsupported claims.
- Using subjective or emotional language.
- Poor organization of ideas.
- Inconsistent terminology.
- Overuse of jargon.
- Poor grammar and spelling.

Characteristics of scholarly writing

Characteristic	What It Means
Evidence-Based	Every claim is supported by data, literature, or scientific evidence.
Clear and Concise	Ideas are communicated using simple, precise language without unnecessary words.
Objective	Avoids personal opinions and emotional language; focuses on facts.
Logical	Information flows in a structured and coherent manner.
Accurate	Findings are reported honestly and precisely.
Transparent	Methods are described in sufficient detail to allow replication.
Critical	Discusses both strengths and limitations of the research.
Ethical	Properly cites sources, avoids plagiarism, and adheres to research ethics.

The Research Publication Journey



Understanding journal article structures

Section	Purpose
Title	Clearly communicates the focus of the study.
Abstract	Provides a concise summary of the study.
Keywords	Improve discoverability in databases.
Introduction	Explains the problem and research question.
Methods	Describes how the study was conducted.
Results	Presents findings without interpretation.
Discussion	Interprets findings and explains their significance.
Conclusion	Summarizes key messages and implications.
Acknowledgements	Recognizes contributions and funding.
References	Lists all cited sources.

IMRAD format

Introduction, Methods, Results, Discussion

IMRAD is the internationally recognized structure for reporting scientific research.

It consists of:

- **I** – Introduction
- **M** – Methods
- **R** – Results
- **D** – Discussion

This structure ensures research is presented logically and consistently.

Section	Purpose	Key Questions Answered
Introduction	Provides background and rationale	Why was the study conducted?
Methods	Explains how the study was carried out	What was done and how?
Results	Presents study findings	What was found?
Discussion	Interprets findings and implications	What do the findings mean?

Research Topic - What did we study?

Topic - It is a clear and concise title stating the subject matter of the research. It captures the what, the where, the who, and in some cases the when.

A GOOD RESEARCH TOPIC IS

01	Comprehensive	fully captures the subject of study
02	Clear	it is not ambiguous and can be understood at first read
03	Concise	should not be more than 21 words
04	Based on logical rationale	built on sound reasoning and evidence
05	It addressed a recognizable problem	responds to an important issue in the field or community
06	Has a clear gap	fills a gap in existing knowledge or practice
07	The researcher has a strong interest in it.	driven by passion and commitment to the topic

The Introduction- Why was the study conducted?

Introduction captures the background, the context, the problem, the objectives and importance of the study

Note: The length of introduction depends on the academic work. Academic papers have shorter introduction while a thesis has a whole chapter dedicated to introduction and background.

KEY ELEMENTS OF THE INTRODUCTION

01

Discuss the topic.

02

Create some context and background.

03

The objectives.

04

Clear gap you intend to capture

05

Explain why your research is important.

COMPONENTS OF THE INTRODUCTION SECTION

The introduction tells the reader why the study matters and what the study aims to do.

1. BACKGROUND TO THE STUDY



Provides context for the study by summarizing previous research.

Starts with what is known and moves to what is not known.

Ends with a research gap that your study will address.

TIPS

- Start globally → region → country → local level.
- Organize by ideas and by years (oldest → newest).
- Use recent literature (last 5 years) unless classics.
- Tell the story clearly and make the gap obvious.

2. STATEMENT OF THE PROBLEM



Presents the problem/gap that your research will address.

Converts a general problem into a specific, well-defined problem.

Includes consequences of not addressing the problem.

Use four parts:

- 1 **Ideal** – How things should be.
- 2 **Reality** – Current situation.
- 3 **Consequences** – What happens if not addressed.
- 4 **Solution** – How the study and findings will help move us closer to the ideal.

3. PURPOSE OF THE STUDY



A clear statement of the overall aim of the study.

Starts with:
“The purpose of this study is to...”

4. OBJECTIVES OR RESEARCH QUESTIONS



Objectives are what you plan to do, broken into smaller parts.

Research questions are used instead of or in addition to objectives.

Tip: Write the objective then turn it into a question.

EXAMPLES

Objective: Identify COVID-19 myths among male youth in Kenya.

Research Question: What are the key COVID-19 myths among male youth in Kenya?

5. JUSTIFICATION FOR THE STUDY



Justification – Why the study is important and how it contributes to filling the gap.

Significance – Who will benefit from the findings and how they will be used.

Rationale – The reason for conducting the study; explains importance and novelty.

6. SCOPE OF THE STUDY



Defines the boundaries and parameters of the study.

Includes population, duration, location, subject matter.

Clearly states what the study will NOT cover.

TIPS

- Scope too broad → not feasible.
- Scope too narrow → poor generalizability.
- State what is included and what is excluded.

TIPS

- Broad objective = your topic in active voice.
- Specific objectives follow stages of learning:

1. Report/Describe – Identify, list, define, state, document
2. Understand – Determine relationships
3. Persuade/Evaluate – Evaluate options

- Use lay language and be direct.



KEY TAKE-HOME MESSAGE

The introduction should lead the reader from the general context to your specific study, show the importance, and clearly explain what the study will do and why it matters.

Methods - How was the Study Conducted?

A well-written Methods section should allow another researcher to replicate your study exactly as it was conducted. Transparency, accuracy, and sufficient detail are the hallmarks of a strong Methods

The Methods Section Should Enable Readers To:

- Understand **exactly how the study was conducted**.
- Assess whether the methods are scientifically appropriate.
- Evaluate the validity and reliability of the findings.
- Replicate the study if needed.
- Determine whether ethical standards were followed

METHODS vs. METHODOLOGY

METHODS	METHODOLOGY
 What did you do?	 Why did you do it this way?
 Describes the procedures used to conduct the study.	 Explains the reasoning behind the chosen research approach.
 Practical and procedural.	 Theoretical and philosophical.
 Included in journal articles .	 Common in theses, dissertations , and research proposals.



KEY TAKE-HOME MESSAGE

Research articles require a Methods section— **not a Methodology section**.
The Methods should provide enough detail for **another researcher to replicate the study**.

Key Components of the Method Section

COMPONENT OF THE METHOD SECTION	WHAT IT SHOULD DESCRIBE	KEY QUESTIONS ANSWERED
1. Study Design	The overall research approach used (e.g., randomized controlled trial, cross-sectional survey, cohort study, qualitative study, mixed methods).	<i>What type of study was conducted?</i>
2. Study Setting	Where and when the study was carried out, including the geographical location, health facilities, schools, communities or institutions involved.	<i>Where and when was the study conducted?</i>
3. Study Participants	The target population, eligibility criteria (inclusion and exclusion), and participant characteristics.	<i>Who participated in the study?</i>
4. Sampling Methods	How participants were selected and the sample size determination. Include sampling technique (random, purposive, convenience, etc.) and sample size calculation where applicable.	<i>How were participants selected?</i>
5. Data Collection	The procedures, tools and instruments used to collect data (e.g., questionnaires, interviews, laboratory tests, observations, focus group discussions).	<i>How were the data collected?</i>
6. Variables and Measurements	The primary and secondary outcomes, exposures, independent and dependent variables, and how each variable was defined and measured.	<i>What was measured and how was it measured?</i>
7. Statistical Analysis	The methods used to analyse the data, including statistical software, tests performed, significance levels and approaches for qualitative analysis where relevant.	<i>How were the data analysed?</i>
8. Ethical Approval	Ethical review committee approval, informed consent procedures, confidentiality measures and permissions obtained before conducting the study.	<i>Was the study conducted ethically?</i>

COMMON MISTAKES IN THE METHODS SECTION

COMMON MISTAKES	WHY IT IS A PROBLEM	BEST PRACTICE
1. Insufficient detail	Readers cannot understand or replicate the study.	Describe each step clearly and completely.
2. Missing study design	Makes it difficult to assess the appropriateness of the methods.	Clearly state the study design in the opening paragraph.
3. Poor description of participants	Readers cannot determine who was included or excluded.	Define the study population and inclusion/exclusion criteria.
4. Inadequate sampling information	Raises questions about bias and representativeness.	Explain the sampling method and sample size determination.
5. Incomplete data collection procedures	Limits confidence in how the data were obtained.	Describe the tools, procedures, timing, and personnel involved.
6. Undefined variables or measurements	Readers cannot interpret the findings correctly.	Clearly define all variables, outcomes, and measurement methods.
7. Weak statistical analysis description	Makes it difficult to evaluate the validity of the analysis.	Specify statistical tests, software used, significance levels, and qualitative analysis methods where applicable.
8. Omitting ethical approval	Raises concerns about research integrity and participant protection.	State the ethics approval committee, approval number (if available), and informed consent procedures.
9. Including results in the Methods section	Blurs the distinction between study procedures and findings.	Describe only what was done—not what was found.
10. Writing in the future tense	Suggests the study has not yet been conducted.	Use the past tense because the study has already been completed.
11. Using vague language	Makes procedures difficult to understand or replicate.	Use precise, specific, and measurable descriptions.
12. Failing to cite established methods	Does not acknowledge previously validated methods.	Cite original sources for standardized or previously published methods.

Results - What did the study find?

The **Results** section presents the findings of your study **without interpretation or discussion**.

- It objectively reports what the data show in relation to your research objectives or questions.
- It answers one simple question: **What did the study find?**

Purpose of the Results Section

- Present the study findings clearly.
- Report the evidence generated from the analysis.
- Summarize key statistical or qualitative findings.
- Lead readers logically toward the Discussion section.

The Results section should present the evidence clearly, objectively, and in the same order as the study objectives without interpretation or opinion.

Common Mistakes in the Results Section

No.	Common Mistake
1.	Interpreting the findings instead of simply reporting them (save interpretation for the Discussion section).
2.	Repeating every value from tables and figures in the text instead of highlighting key findings.
3.	Presenting results out of order, making it difficult to follow the research objectives.
4.	Reporting irrelevant or excessive statistics that do not address the study objectives.
5.	Failing to reference tables and figures within the narrative.
6.	Using unclear or poorly labelled tables and graphs, making findings difficult to interpret.
7.	Selective reporting of results, omitting non-significant or unexpected findings.
8.	Including opinions, recommendations, or conclusions, which belong in the Discussion and Conclusion sections.

COMPONENTS OF THE RESULTS SECTION

Component	What It Includes
Participant Characteristics	Response rate, demographics, baseline characteristics
Descriptive Statistics	Frequencies, percentages, means, medians, standard deviations
Main Findings	Results addressing each study objective
Inferential Statistics	Hypothesis tests, confidence intervals, regression analysis
Tables	Numerical summaries of findings
Figures	Graphs, charts, maps and diagrams
Qualitative Findings	Themes, categories and participant quotations (where applicable)
Summary of Key Findings	Brief transition to the Discussion

Discussions - Why do these findings matter?

The Discussion explains the meaning and importance of the study findings. It interprets the results, compares them with previous research, discusses their implications, acknowledges limitations, and highlights the contribution of the study.

It answers one key question: What do these findings mean?

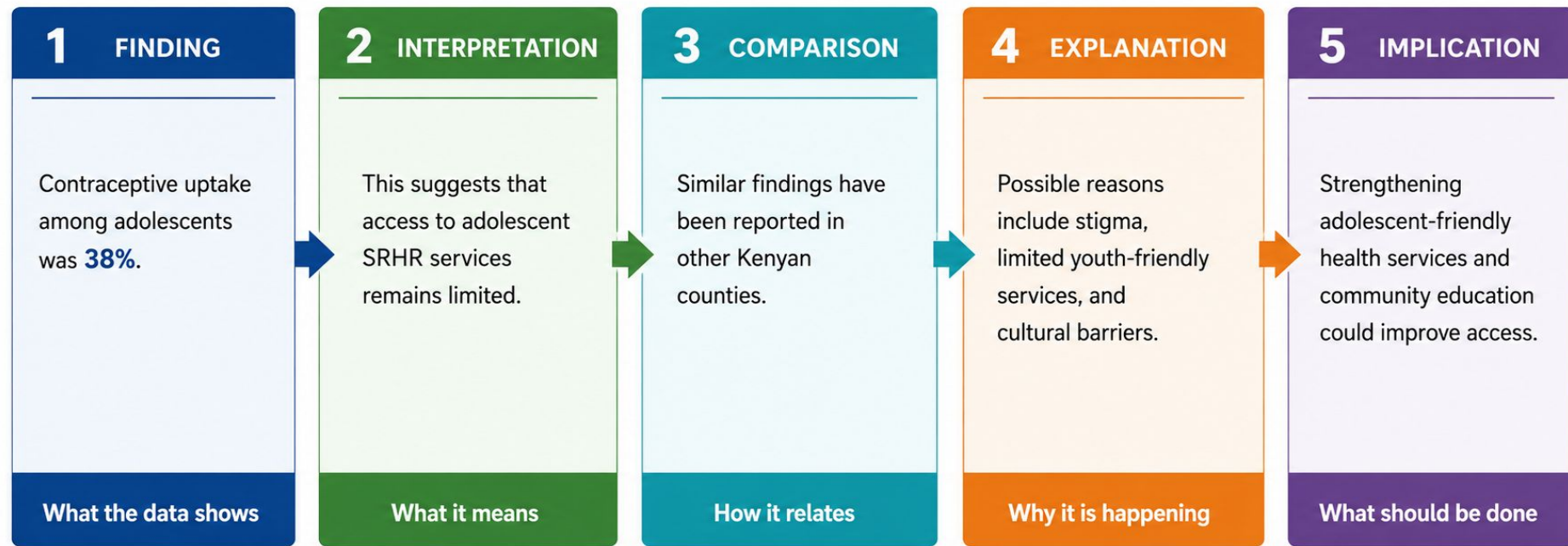
The "4 Cs" of a Strong Discussion

Every Discussion Should:

1. **Connect:** Connect your findings to your research objectives.
2. **Compare:** Compare your findings with previous studies.
3. **Critically Explain:** Explain why the findings occurred.
4. **Conclude:** Describe the implications and significance of the findings.

Writing a good discussion section

The Results section tells readers what you found. The Discussion section explains what those findings mean, why they matter, and how they contribute to existing knowledge



COMPONENTS OF THE DISCUSSION SECTION

COMPONENT	PURPOSE
 1. Summary of Key Findings	Briefly highlight the most important findings without repeating all the results.
 2. Interpretation of Findings	Explain what the findings mean and why they matter.
 3. Comparison with Previous Studies	Compare findings with existing literature and explain similarities or differences.
 4. Possible Explanations	Discuss biological, behavioural, social, policy or contextual reasons for the findings.
 5. Strengths of the Study	Highlight aspects that increase confidence in the findings.
 6. Study Limitations	Acknowledge weaknesses and their possible impact on interpretation.
 7. Implications	Explain what the findings mean for policy, practice, programmes or future research.
 8. Conclusion/ Transition	End with the key take-home message leading into the Conclusion section.

COMMON MISTAKES IN THE DISCUSSION SECTION

COMMON MISTAKE	BEST PRACTICE
Repeating the Results section	Interpret the findings instead of restating them.
Ignoring previous literature	Compare findings with relevant studies.
Overstating conclusions	Base conclusions only on the evidence.
Ignoring unexpected findings	Discuss and explain possible reasons.
Hiding study limitations	Acknowledge limitations honestly.
Making unsupported recommendations	Ensure recommendations are grounded in the findings.
Introducing new results	Discuss only findings already presented in the Results section.

Session Discussion



What are some of the challenges you have experienced when building the Introduction section of your Thesis or your Research Article?

A man with dark hair, wearing a blue shirt, is shown from the chest up, looking slightly to the right with a thoughtful expression. The background is blurred, showing what appears to be a store or office setting with shelves and lights. Overlaid on the image are three text boxes. The first box, at the top, contains the text 'Abstract vs. Introduction:' in red. The second box, in the middle, contains 'Do You Know the' in black. The third box, at the bottom, contains 'Difference?' in black. The text boxes are white with black borders. The overall image has a soft, professional feel.

Abstract vs. Introduction:

Do You Know the

Difference?

Abstract vs Introduction

An **abstract** and an **introduction** serve different purposes, even though they both appear at the beginning of a document. The key difference is that an **abstract summarizes the entire document**, while an **introduction sets the stage for the discussion that follows**.

Abstract	Introduction
Provides a concise summary of the entire document.	Introduces the topic and explains why it is important.
Enables readers to quickly determine whether the document is relevant to them.	Provides background and context to help readers understand the issue.
Summarizes the problem, purpose, methods (if applicable), key findings or messages, and conclusions or recommendations.	Describes the problem, context, rationale and objectives, but does not summarize the entire document.
Written after the document is completed, although it appears first.	Written at the beginning of the document development process and refined as needed.
Usually one paragraph (150–300 words, depending on the document).	Can range from one paragraph to several pages, depending on the document type.
Should be able to stand alone without reading the rest of the document.	Leads readers into the main body of the document and relies on subsequent sections for details.

Questions



Session 2:

Building a Strong Manuscript



Topics

01

Writing effective abstracts

02

Literature review development

03

Constructing arguments

04

Presenting methods and results